TROUBLESHOOT-RTP-007: Tablet Chipping and Edge Defects

Machine: RTP-7
Fault Type: Product quality defect
Severity: High — batch may need to be quarantined

What the Technician Observes

The technician typically observes tablets with visible chipping along the edges, particularly around the circumference or at embossed regions. The edges may appear rough, broken, or flaked rather than smooth and well-defined. In some cases, minor lamination or capping may accompany the defect. During operation, the machine may produce a slightly irregular sound, such as intermittent knocking or scraping, indicating possible tooling wear or misalignment. Sensor readings may show fluctuations in compression force (±8–10% instead of normal ±5%), slight variations in turret speed stability, or inconsistent fill depth. Additionally, increased dust generation and powder accumulation near the feeder or die table may be noticed.

Possible Causes

Cause 1 — Worn or Damaged Punch Tips (Most likely — ~60% of cases)

Over time, upper and lower punches are subjected to continuous mechanical stress, leading to gradual wear of the punch tips, especially at edges and embossing features. This wear reduces the sharpness and integrity of the punch face, causing improper compaction and weak tablet edges. Micro-chipping on punch tips can also transfer directly to the tablet surface, resulting in visible defects. If not replaced at regular intervals, worn punches can create uneven pressure distribution during compression, exacerbating edge damage. This issue is more common in high-speed production or abrasive formulations.

How to confirm: Remove and visually inspect punches under adequate lighting or magnification. Look for rounding, scratches, or micro-chips on the punch tips. Compare with a new punch profile if available.

Cause 2 — Improper Compression Force or Dwell Time (~25% of cases)

Incorrect compression settings, particularly excessive compression force or insufficient dwell time, can lead to brittle tablets that chip easily during ejection. High compression force can create internal तनाव (stress) within the tablet, making edges fragile. Conversely, low dwell time due to high turret speed may not allow sufficient bonding of particles, leading to weak संरचना. Over time, this imbalance results in tablets that cannot withstand mechanical handling, causing edge defects.

How to confirm: Review machine HMI logs for compression force trends and turret speed. Compare with validated process parameters. Conduct a hardness test; excessively hard or brittle tablets indicate over-compression.

Cause 3 — Granule Quality Issues (~15% of cases)

Poor granule properties, such as low moisture content, improper particle size distribution, or inadequate binder levels, can lead to weak tablet संरचना. Dry or over-milled granules lack पर्याप्त cohesion, causing edges to break during compression or ejection. This issue often develops due to upstream process deviations in granulation or drying stages.

How to confirm: Check granule moisture content (target typically 1.5–3%). Perform sieve analysis for particle size distribution. Review batch manufacturing records for granulation parameters.

Diagnostic Steps

Step 1 — Inspect Tablet Samples

Begin by collecting tablet samples from different stages of production, including initial, mid, and current output. Examine them visually and under magnification to identify the severity and pattern of chipping. Uniform defects across all tablets suggest a systemic issue, whereas random defects may indicate localized tooling damage. Compare with approved reference samples.

Step 2 — Check Punch Condition

Stop the machine following LOTO procedure and remove a representative set of punches. Inspect both upper and lower punches for wear, corrosion, or damage. Pay special attention to edges and embossed areas. A punch tip radius exceeding 0.05 mm from original specification is considered worn. Document findings and segregate defective punches.

Step 3 — Review Compression Parameters

Access the machine control panel and review compression force, pre-compression settings, and turret speed. Normal compression force should be within validated range (e.g., 10–25 kN depending on product). Any deviation beyond ±5% should be investigated. Check dwell time indirectly via turret speed; excessively high speeds reduce dwell time.

Step 4 — Evaluate Granule Properties

Collect granule samples and test for moisture content using a moisture analyzer. Ideal range is typically 1.5–3%. Perform sieve analysis to ensure proper particle size distribution. Excess fines (<10%) or oversized granules can affect compaction quality.

Step 5 — Inspect Die and Ejection System

Examine dies for wear, scratches, or improper alignment. Check ejection cam and take-off blade for smooth operation. Misalignment or excessive friction during ejection can cause mechanical stress leading to chipping. Ensure die bore clearance is within 0.02–0.05 mm.

Step 6 — Monitor Machine Operation

Run the machine at low speed and observe punch movement and compression cycle. Listen for abnormal sounds such as knocking or vibration. Use vibration meter if available; readings above 5 mm/s indicate mechanical issues.

Step 7 — Check Environmental Conditions

Verify humidity and temperature in the compression room. Low humidity (<30% RH) can dry out granules and increase brittleness. Ensure HVAC system is functioning within GMP limits.

Corrective Actions

If cause is Worn or Damaged Punch Tips:

Replace all worn punches with new, qualified tooling from approved inventory. Ensure punches are from the same batch and meet dimensional specifications. Apply proper lubrication and install using calibrated torque (18–22 Nm). Estimated downtime: 2–4 hours depending on turret size. Perform trial run and verify tablet quality before resuming production.

If cause is Improper Compression Force or Dwell Time:

Adjust compression force and turret speed to validated parameters. Reduce compression force gradually while monitoring tablet hardness and friability. If dwell time is insufficient, decrease turret speed by 10–15% to allow better particle bonding. No parts required. Estimated downtime: 30–60 minutes including validation checks.

If cause is Granule Quality Issues:

Hold production and inform production and QA teams. Reprocess or adjust granules by adding binder solution or blending with fresh batch if permitted. Ensure moisture content and particle size distribution meet specifications before restarting compression. Estimated downtime: 4–8 hours depending on reprocessing.

When to Escalate

Escalate to the maintenance supervisor if punch wear is excessive across multiple sets within a short period, indicating systemic issues. Contact OEM support if turret misalignment, die table damage, or persistent vibration is observed. Involve QA immediately if multiple batches are affected or if defects exceed acceptable limits.

Batch Disposition

Segregate all tablets produced during the defect period and label as “Under Investigation.” Conduct quality testing including hardness, friability, and dissolution. Based on QA assessment, the batch may be reprocessed, partially rejected, or fully discarded. Ensure complete documentation for GMP compliance and audit readiness.
